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Industrial Internet of Things (IIoT) and industrial robots for smart manufacturing are a crucial part of Industry 4.0 and are responsible for the transformation. Industry 4.0 market is likely to reach US\$ 156.6 billion from 2019 to 2024, expanding at a CAGR of 16.9 per cent, according to a report from Markets and Markets.

The impact is that data-optimised smart machines are connected to the entire supply chain so that manufacturing is in sync with the varying requirements and conditions in real time. Existing technologies like supervisory control and data acquisition (SCADA) systems and programmable logic controllers (PLCs) are also changing. Machines use sensors to collect comprehensive, real-time data that is crucial for decision-making using AI. Transparency and security are improved as data for tracking a product's lifecycle gets stored in the blockchain. Increased use of 3D printing has resulted in faster time-to-market for many products. Even before the product reaches the market, technologies like augmented reality (AR) and virtual reality (VR) can be used for testing it. By deploying a 4G/5G network for wireless connectivity, processing of large volumes of data has become possible, and reconfiguration of the production line has become easier.

All these modifications eliminate inventory issues and significantly decrease production waste. In this article we discuss how customers are at the forefront of this revolution by being responsible for driving the creation of products of the future.

Innovative solutions for an enhanced experience

Manufacturers are increasingly integrating existing equipment with smart sensors to enable data collection in real time, which can be used for predictive maintenance and cloud-based AI. This saves consumers from inconvenience in the future. Germany-based ThyssenKrupp uses sensors to connect its elevators to the cloud. Data collected from the sensors is utilised for predicting when maintenance is required before the elevators break down.

Traditional mass manufacturing cannot fulfil the growing demand for customisation and personalisation for individual customers. This is further fuelling the growth of innovative solutions. For instance, US-based Local Motors realised that small factories can solve this issue. The company that earned limelight when in 2016 it unveiled AI-powered self-driving shuttle bus Olli focuses on low-volume manufacturing of open source motor vehicle designs and operates multiple micro-factories.

In 2018, Alibaba introduced G Plus, an unmanned logistics vehicle, which uses solid-state light-detection and ranging (lidar) technology to deliver goods purchased online to customers. It can carry several packages of different sizes and has a storage locker with facial recognition to keep food warm. This was done